Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/12/2025
PI Name:	Delp, Michael Ph.D.		
Project Title:	Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	08/01/2019	**End Date:**	07/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	End date changed to 07/31/2025 per F. Hernandez/ARC and NSSC information (Ed., 12/23/24) End date changed to 07/31/2024 per F. Hernandez/ARC (Ed., 6/17/24) End date changed to 01/31/2024 per NSSC information (Ed., 6/22/23) End date changed to 07/31/2023 per NSSC information (Ed., 8/25/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Xiao Wen M.D. (Loma Linda University) Muller-Delp, Judy Ph.D. (Florida State University) Shirazi-Fard, Yasaman Ph.D. (NASA Ames Research Center) Ledermann, Thomas Ph.D. (Florida State University)		
Grant/Contract No.:	80NSSC19K1322		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	This proposal is one of two proposals selected for funding in response to solicitation NNJ16ZSA001N-SRHHC “Topic 6: Combined Effects of Simulated Weightlessness and Space Radiation on Cardiovascular Structure and Function,” and then descoped and merged at the direction of NASA with a second selected proposal (PI John Baker, PhD; Defining the Relationship Between Simulated Weightlessness and Space Radiation on Cardiovascular Disease, and the Effectiveness of a Countermeasure).” Emerging research demonstrates that low-dose (<0.5 Gy) environmental and medical radiation exposure increases the risk of mortality due to ischemic coronary artery (heart attack) and cerebrovascular (stroke) disease. Results from a recent study of mortality rates among non-flight, low Earth orbit, and Apollo lunar astronauts also suggests, but does not prove, that Apollo lunar astronauts have a higher proportional mortality rate due to cardiovascular disease than non-flight astronauts. Despite the potential for low-dose radiation exposure to have substantial adverse effects on the cardiovascular system, little research has been conducted to investigate this phenomenon. This is particularly urgent as groups of nations and corporate entities are actively planning manned missions back into deep space (i.e., the Moon and Mars), where exposure to galactic comic radiation is pervasive. The present proposal seeks to address some of the fundamental questions that remain outstanding, including: 1) at what dose does simulated space radiation adversely affect cardiac and cerebral vascular function? 2) Does space radiation and weightlessness interact to produce greater coronary and cerebral vascular dysfunction than either alone? 3) Does space radiation and weightlessness have differential effects on coronary and cerebral arteries? 4) Is the cardiovascular dysfunction induced by space radiation and weightless sustained over a long period of recovery, potentially leading to chronic morbidity and mortality? 5) Does dietary intervention (e.g., blackberry supplement) that normally has positive impacts on the cardiovascular system serve as effective countermeasures for radiation-induced cardiovascular dysfunction, and if so, is it effective in both the coronary and cerebral circulations? 6) Is the putative cardiovascular dysfunction induced by space radiation and weightlessness similar in males and females, and is a dietary countermeasure equally effective in both sexes? Results from these studies will lead to a greater understanding of the sustained cardiovascular health risks for prolonged deep space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cardiovascular disease risk due to radiation exposure has been changing over the past several decades, with greater understanding that cardiovascular disease risk is elevated with increasingly lower doses of exposure. The current project will add significantly to a very small body of literature on the long-term consequences of low-dose irradiation on the function of cerebral and coronary vascular health, which has a major impact on the risk of stroke and coronary artery disease in humans.
Task Progress:	For this reporting period, the major activities have been to complete the immunohistochemical analysis of jugular vein structure and signaling molecule expression, initiate the analysis of the cervical lymphatic physiological responses, and section all mesenteric artery, vein and lymphatic vessels in preparation for structural analysis and determination of signaling molecule expression. Results for the jugular vein demonstrate that simulated deep space irradiation alters physiological venomotor responses, even after a very long recovery period post-irradiation. Additionally, perhaps some of the most significant findings from preliminary analysis of the cervical lymphatic vessels draining the head indicate that both simulated space radiation and weightlessness alter the capacity of lymphatic vessels to pump fluid. These observations with both the jugular vein and cervical lymphatic vessels could indicate long-term problems with both immune function and fluid pressure regulation in the head and eyes following flight into deep space.
Bibliography Type:	Description: (Last Updated: 07/09/2025)